Assignment 4

1 Pets Expression Classification

In this assignment you will be provided with a dataset that contains 1000 face images of various pets, such as dogs, cats, rabbits, hamsters, sheep, horses, and birds. The images capture the diversity of expressions these animals can display, such as happiness, sadness, anger etc. The dataset can be used to train and evaluate models that can recognize the emotions of pets from their facial expressions. This can help pet owners understand their pets better and improve their well-being.

2 Building Well-Known Networks

You will start by building the following well known networks from scratch using Keras or PyTorch:

- Visual Geometry Group (VGG)
- Residual Neural Network (ResNet)
- MobileNet
- Inception V3
- DenseNet121

Report The accuracy and mean Average: Precision, Recall, F1-score, and the confusion matrix produced by each model.
You are required to use data augmentation in the training process.

3 Transfer Learning

Instead of training the models staring from random state, we can use transfer learning to start with a better set of networks' weights trained on famous datasets. Here, you will use ImageNet weights to enhance the training of well-known networks. Choose the best performed network from the previous part and apply transfer learning with ImageNet weights. In this part use the networks models built-in Keras or PyTorch. Report the new performance of the network.

Good Luck!

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4 Notes

- You are required to deliver your well-commented code, and a report containing all the used networks and models, with their obtained results.
- You can work in groups of 3.

5 References

- Pet's Facial Expression Image Dataset